

U.S. COCOA IMPORT & PROCESSING

Business case and marginal-cost model for large-scale cocoa-bean imports into the United States

INVESTMENT QUESTION

Can a vertically integrated U.S. cocoa platform create attractive economics by importing beans at scale, processing domestically, and capturing value through industrial chocolate and premium consumer products?

	Reference / model value	Interpretation
U.S. cocoa-bean imports, 2024	~198,000 t	Sharp decline from 2000-22 average
2000-22 average imports	~425,000 t/year	Indicative historical scale
Illustrative target throughput	100,000 t/year	Large regional/national platform
Illustrative cocoa price range	\$2,500-\$12,000/t	Sensitivity range, not a forecast
Illustrative bar	70 g, 70% cocoa	Basis for marginal-cost model

Sources: USDA Economic Research Service; U.S. International Trade Commission tariff schedule; International Cocoa Organization. See source notes on page 8.

1. Executive Summary

There is a credible industrial case for increasing direct U.S. imports of cocoa beans, provided the strategy extends beyond commodity trading into domestic processing and downstream chocolate manufacture. USDA reports that U.S. cocoa-bean imports fell from roughly 425,000 tonnes per year during 2000-22 to about 198,000 tonnes in 2024 as cocoa prices surged. The United States nevertheless remains a very large chocolate-consuming market.

The investment thesis is therefore not simply "buy cheap beans and sell chocolate." It is a vertical cocoa platform: direct origin procurement, ocean logistics, U.S. roasting/grinding, liquor/butter/powder recovery, industrial chocolate manufacture, and selectively owned consumer brands.

The core modelling result is that the raw-bean price has a surprisingly modest marginal effect on the factory cost of a premium chocolate bar when expressed per bar. Under an illustrative 70 g, 70%-cocoa formulation and an 80% bean-to-usable-cocoa conversion assumption, each \$1,000/t increase in bean price adds about \$0.061 per bar.

That resilience is strategically useful: the business can operate across a wide cocoa-price range if it captures sufficient value from processing, co-products and downstream margin.

2. Market and Supply Context

USDA data show that U.S. cocoa-bean imports averaged approximately 425,000 tonnes annually over 2000-22, but fell to around 198,000 tonnes in 2024. USDA attributes the contraction to the exceptional rise in cocoa prices. The 2024 U.S. bean-import bill was approximately \$1.11 billion, according to U.S. Department of Labor research citing trade data.

	Volume
2000-22 average	~425,000 t/year
2024	~198,000 t
Illustrative platform - Phase 1	25,000-50,000 t/year
Illustrative platform - scale	100,000 t/year
Illustrative national scale	200,000 t/year

The historical data should be treated as a scale benchmark, not as an assertion that U.S. imports must immediately return to 425,000 tonnes. A new entrant could establish a regional foothold and scale as processing and customer contracts mature.

3. Procurement Strategy

A scalable U.S. cocoa platform should diversify across producing origins rather than depend on a single country or commodity specification. Ecuador, Côte d'Ivoire and the Dominican Republic are among the major suppliers to the U.S. market in recent trade data. A procurement portfolio can separate commodity-grade volume from fine-flavour and specialty cocoa.

Procurement tier	Purpose	Commercial role
Commodity cocoa	Scale and cost	Mass-market chocolate and industrial applications
Premium cocoa	Differentiation	Higher-value chocolate and couverture
Single-origin / specialty	Brand value	Premium and luxury bars
Long-term contracted supply	Security	Volume planning and traceability

4. Value-Chain Architecture

The strongest strategic configuration is an integrated value chain in which the U.S. facility receives beans and converts them into multiple saleable outputs. This reduces reliance on any single end-product margin.

Farm / cooperative → ocean logistics → U.S. port → warehouse → roast / winnow / grind → liquor + butter + powder → industrial chocolate → branded products

The platform can monetise the bean through cocoa liquor, cocoa butter, cocoa powder, bulk chocolate and finished consumer products. Co-product economics should be explicitly modelled rather than treating all bean value as a cost absorbed by the chocolate bar.

5. Cocoa Bean Price Sensitivity

Cocoa markets have been unusually volatile. ICCO reported a 2024/25 annual average indicator price of roughly \$6,612/t and substantial subsequent volatility. The following sensitivity range is therefore deliberately broad and should be treated as a scenario framework, not a price forecast.

\$2,500	\$0.153	\$0.613
\$3,000	\$0.184	\$0.644
\$4,000	\$0.245	\$0.705
\$5,000	\$0.306	\$0.766
\$6,000	\$0.367	\$0.828
\$7,000	\$0.429	\$0.889
\$8,000	\$0.490	\$0.950
\$10,000	\$0.613	\$1.073
\$12,000	\$0.735	\$1.195

Model assumptions: 70 g bar; 70% cocoa share; 80% conversion from purchased beans to usable cocoa material; \$0.46/bar of non-bean factory cost. These are illustrative engineering assumptions and must be replaced with actual plant yields, formulation, packaging and labour quotes.

6. Marginal Cost Model

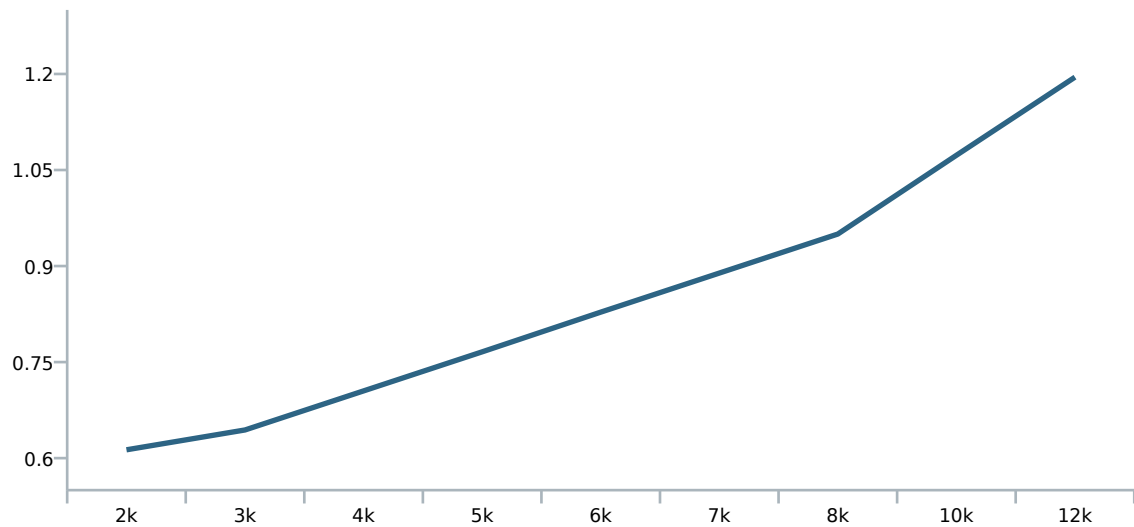
For the illustrative bar, the cocoa requirement is 49 g of usable cocoa material. With an assumed 80% conversion efficiency from whole beans, required purchased beans are 61.25 g per bar.

The resulting marginal cocoa cost is approximately:

Cocoa cost per bar = 0.06125 kg × cocoa price (\$/kg)

Thus, the marginal cost with respect to the cocoa bean price is approximately \$0.06125 per \$1,000/t. Put differently, a \$4,000/t move in cocoa price changes the illustrative bar cost by about \$0.245.

7. Marginal Factory Cost Curve



Under the stated assumptions, the relationship is approximately linear. The business therefore needs to manage cocoa price risk, but the raw bean component does not overwhelm the economics of premium chocolate at plausible commodity-price levels.

8. Product and Margin Architecture

The downstream portfolio should be deliberately diversified. A processing plant can sell intermediate products while using a portion of output to manufacture branded chocolate. This reduces dependence on retail brand-building alone and creates a B2B floor for plant utilisation.

Business layer	Products	Primary economic function
Cocoa trading	Whole beans	Procurement and origin access
Primary processing	Liquor / mass	Industrial ingredient margin
Co-product recovery	Butter / powder	Monetise full bean value
Industrial chocolate	Couverture / bulk chocolate	B2B volume and plant utilisation
Consumer chocolate	Premium bars / boxes	Higher downstream margin and brand equity

9. What 100,000 Tonnes Means

At 100,000 tonnes of annual bean throughput, every \$1,000/t movement in procurement price changes annual raw-bean spend by approximately \$100 million. At \$5,000/t, annual bean procurement would therefore be about \$500 million, before freight, insurance, financing and processing.

Throughput (t)	Price (\$/t)	Annual Spend (\$m)
25,000 t	\$5,000/t	\$125m
50,000 t	\$5,000/t	\$250m
100,000 t	\$5,000/t	\$500m
100,000 t	\$6,000/t	\$600m
200,000 t	\$6,000/t	\$1.2bn

At the illustrative 16,327 bar-equivalents per tonne, 100,000 tonnes corresponds to roughly 1.63 billion 70 g bar-equivalents. Actual commercial output would be distributed among chocolate, liquor, butter, powder and other products, so this is a scale indicator rather than a production forecast.

10. Strategic Case for U.S. Processing

A potentially attractive structure is to import beans rather than finished chocolate and perform more processing inside the United States. The U.S. tariff schedule lists cocoa beans under HTS 1801.00.00 as Free. Processed cocoa products are classified separately, creating a rationale for examining where value-added processing should occur in the supply chain.

Domestic processing can also improve traceability, create shorter physical supply chains to U.S. manufacturers, and provide a platform for higher-value branded products. However, the trade case must be assessed on a product-by-product tariff basis; this report does not treat tariff treatment as a blanket conclusion for every downstream cocoa product.

11. Principal Risks

Risk	Why it matters	Mitigation
Cocoa-price volatility	Working-capital and margin pressure	Long-term contracts, hedging, diversified origins
Yield / conversion	Actual plant yield may differ from 80% assumption	Pilot trials and validated mass balances
Origin concentration	Weather, disease and political risk	Multi-origin procurement portfolio
Energy and logistics	Processing economics depend on power and freight	Site selection, long-term energy and freight contracts
Brand execution	Consumer margin requires marketing and distribution	Retain B2B outlet for industrial volume
Capital intensity	Processing and chocolate plants require material capex	Phased build or acquisition strategy

12. Recommended Business Model

The strongest commercial structure is a four-layer platform:

1. Cocoa procurement. Direct relationships with farms, cooperatives and exporters; origin diversification; quality specifications and traceability.
2. Primary processing. U.S. roasting, winnowing, grinding and pressing, with explicit co-product monetisation.
3. Industrial chocolate. Supply bakeries, food manufacturers, foodservice and private-label customers.
4. Consumer brands. Use selected production to build premium dark, milk, single-origin and specialty chocolate lines.

This structure reduces the strategic dependence on consumer-brand success by maintaining a B2B channel that can absorb industrial output.

13. Conclusion

The U.S. cocoa opportunity is best understood as an infrastructure and processing opportunity rather than a simple chocolate-brand opportunity. The U.S. market has a demonstrated history of absorbing substantial volumes of cocoa beans, while 2024 imports fell to around 198,000 tonnes during an extraordinary commodity-price shock. A well-capitalised entrant could use direct sourcing and domestic processing to rebuild part of that supply chain and capture value at multiple stages.

The marginal-cost model provides an encouraging initial signal: with the stated 70 g / 70% cocoa formulation, a \$1,000/t increase in bean prices adds only about \$0.061 per bar to the illustrative factory cost. The more important variables for investment analysis are therefore plant yield, co-product realisation, energy, freight, labour, packaging, capacity utilisation, working capital and downstream gross margin.

The recommended next step is a full project-finance model with origin-specific bean prices, freight lanes, plant mass balances, equipment quotations, energy assumptions, co-product prices, customer contracts, working-capital requirements and realistic chocolate conversion yields. That model should be run across commodity-price and demand scenarios before capital is committed.

14. Sources and Model Notes

USDA Economic Research Service, Charts of Note: U.S. cocoa-bean imports and the impact of elevated cocoa prices. <https://ers.usda.gov/data-products/charts-of-note/110921>

U.S. Department of Labor, Cocoa Supply Chain report, including 2024 U.S. cocoa trade data and major suppliers. https://www.dol.gov/sites/dolgov/files/ILAB/Cocoa%20supply%20chain_final%20report.pdf

U.S. International Trade Commission, Harmonized Tariff Schedule, HTS 1801.00.00. <https://hts.usitc.gov/>

International Cocoa Organization, Annual Report 2024/25 and cocoa market reports, including indicator-price data and production estimates. <https://www.icco.org/statistics/> and <https://www.icco.org/wp-content/uploads/ICCO-Annual-Report-2024-2025-EN.pdf>

Model note: The bar-cost model is intentionally illustrative. It uses 61.25 g of purchased beans per 70 g bar, equivalent to a 70% cocoa share and 80% bean-to-usable-cocoa conversion. The non-bean factory-cost allowance of \$0.46/bar is a planning assumption, not a market quotation. Freight, duties on downstream products, financing, co-product credits, plant depreciation, distribution, marketing, retailer margins, taxes and corporate overhead are not included unless explicitly stated. The model should not be used as a final investment decision without plant-specific engineering and commercial validation.

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This report is a business-case framework, not an offer of securities or financial advice. All investment decisions should be based on independently verified technical, commercial, legal, regulatory and financial due diligence.